Technical reference manual
1.0.0

EtherBench

Extension boards

The whole project is based on different boards, including an extension board. The latter one target different missions:

- Adapt the signal to the required usage (voltage, isolation, current...).
- Protect the main board, in case of failure of the DUT (short-circuit, over-voltage...).
- Be a devboard by itself, in the meaning, nothing prevent (that's even an handled case !) to put the target on the devboard, and play with it !

Therefore, it must be replacable, but also physically tied to the main board to protect them against mechanical damages. To ensure that, we choosed to use the PCIe 4x connector, to provide both power and signals to this board. The form factor isn't compliant, as well as the pinout.

This connector was preferred, as it's able to use standard 1.6mm PCB, which only require fingers and an angle. This greatly reduce cost for custom boards, as it won't require any costly option. From our checks, you can get a compatible board from JLCPCB starting at 25€ (4 layers + ENIG).

Mechanical format

The board require to be in this format. The length that goes outside (L) can be extended beyond, at the condition:

- Your case can accept it
- The mechanical structure remain adequate for the load to be applied.

[TODO : Export from altium]

Electrical consideration

There's two power rails :

Rail	Value	Current	Host capacitance	Notes
MASTER_VDD	Fixed to 3.3V	Up to 0.5 A	At least 10 uF	Used to supply the onboard EEPROM and level shifters
DUT_VDD	Variable between 0.6 and VUSB	Up to 4 A	At least 100 uF	Used to supply the test logic

All signals that are going over the PCIe connector are referrenced in the MASTER_VDD domain. DUT_VDD is only applicable after the outputs are adapted by buffer, or any logic circuits.

Pinout

The PCIe pinout is inspired by the standard PCIe, but vastly differ passed the notch. In any case, **DO NOT HOOK STANDARD PCIe BOARDS** (The layout make that impossible, but with extenders...).

Here the complete pinout :

Pin	Side A	Side B
1	BOARD_PRESENT	DUT_VDD
2	DUT_VDD	DUT_VDD
3	DUT_VDD	DUT_VDD
4	GND	GND
5	TCK / SWCLK	MANAGEMENT_SCL
6	TDI	MANAGEMENT_SDA
7	TDO / SWO	GND
8	TMS / SWDIO	MASTER_VDD
9	DUT_VDD	DUT_RESET
10	DUT_VDD	GND
11	BUFFER_EN	DUT_PRESENT (DUT must pull to DUT_VDD)
NOTCH	**NOTCH**	**NOTCH**
12	I3C_SDA	I3C_SCL
13	CLK	GND
14	GND	USB DP
15	GPIO PUSH_PULL 0	USB DM
16	DAC 0	GPIO OPEN_DRAIN 0

17	DAC 1	GPIO OPEN_DRAIN 1
18	GND	GND
19	GPIO PUSH_PULL 1	CAN TX
20	GND	CAN RX
21	ADC 0	GPIO OPEN_DRAIN READ 0
22	ADC 1	GPIO OPEN_DRAIN READ 1
23	SPI SS1	SPI SS0
24	GND	SPI_MOSI
25	SPI_MISO	GND
26	SPI SCLK	GND
27	USART1 RTS	USART1 RX
28	USART1 CTS	USART1 TX
29	USART2 RX	GND
30	USART2 TX	GND
31	GND	USART3 RX
32	GND	USART3 TX

Isolation

The PCIe slot isn't isolated from the DUT by default. This remains an option for specific required. In that case, an isolated power supply will also be required to convey the provided voltage to the isolated island.

The motherboard was designed with that requirement in mind, therefore only the two support screw require a carefull attention, as they're on the main ground domain. It's on the daughter board to treat them correctly.